

PRODUCT CASE STUDY · EXECUTIVE PRD

BuildOS

An Intelligence Platform for Home Construction

BuildOS connects the homeowners, architects, contractors, suppliers and institutions of a residential build into one shared source of truth — turning a fragmented, WhatsApp-and-spreadsheet process into a coordinated, transparent one.

THE CORE INSIGHT

“The process is not lacking expertise; it is lacking integration.”

Prepared by

Vijetha Nissankara

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AI Product Management Portfolio Case Study

SECTION 01

Executive Summary

The problem

Residential construction in India is a 12–24 month, high-stakes project that homeowners run themselves — coordinating architects, engineers, contractors, suppliers and banks across WhatsApp threads, spreadsheets, calls and paper records, with no construction expertise and no single source of truth.

Who feels it

The homeowner — typically a working professional funding and overseeing their one home build — becomes an accidental project manager. A secondary set of professionals (contractors, architects) also lose time to the same fragmentation from the other side of the table.

The MVP

A homeowner-facing project workspace — stakeholders, quotations, budget tracking, documents, tasks and a status dashboard — with a contractor-facing view for the same shared data. AI is introduced only where enough structured context already exists (quotation comparison, budget-deviation flags), not as a headline feature.

Where AI adds value

AI turns scattered inputs (a quotation PDF, a WhatsApp update, a budget line) into a structured recommendation a homeowner can act on — with a human always approving the consequential step. Over time, targeted assistance can extend into agentic workflows such as vendor follow-ups, always with human approval retained for anything binding.

The long-term opportunity

India's individual-home construction market is large but structurally under-digitized (85%+ unorganized, per industry disclosures). A platform that earns trust on one home build has a natural path into the surrounding ecosystem — suppliers, financial institutions, and professional networks — turning a one-time transaction into a durable data and trust layer for residential construction.

What BuildOS is

A coordination platform that brings project information, stakeholders, decisions, quotations, budgets, documents and progress into one connected workspace — with an AI layer that surfaces what needs attention, not another chat tool or a replacement for expertise.

Why it's different

BuildOS doesn't perform construction or replace the professionals who do. It sits above the existing team as connective tissue — “coordinate your own chosen team,” not a heavier full-service alternative.

STRATEGIC TAKEAWAYS

- Fragmentation, not lack of expertise, is the problem worth solving — this reframes BuildOS as connective infrastructure, not a construction company.
- The MVP wins by being narrow: a shared workspace plus quotation and budget clarity, not a feature-complete platform on day one.
- AI should follow context, not precede it — recommendation quality depends on the structured data the MVP collects first.
- The market is real but modest at Year-3 scale (₹2.2 crore base-case SOM); the bigger prize is the ecosystem position a trusted coordination layer can earn over time.

SECTION 02

Problem & Opportunity

Today: a fragmented construction journey

A homeowner sits at the center of a project they did not train to run, connected to every stakeholder through a different, disconnected channel:

STAKEHOLDERS

- Architect
- Structural engineer
- Contractor
- Material suppliers
- Consultants
- Bank / lender
- Local authorities

CHANNELS IN USE

- WhatsApp threads
- Phone calls
- Spreadsheets
- Email
- Site photos
- Paper records / physical files

What this produces

Symptom	What it looks like for the homeowner
Information fragmentation	No single place holds the current quotation, the approved budget, and the latest site photo together.
Coordination overhead	The homeowner manually relays updates between architect, contractor and supplier.
Decision uncertainty	Material and vendor decisions are made without a clear comparison of cost, quality and timeline trade-offs.
Budget opacity	Spend-to-date and remaining budget are reconstructed from receipts and memory, not tracked live.
Timeline uncertainty	Delays surface only after they've already affected the next stage of work.
Repetitive follow-ups	The homeowner chases the same vendor for the same pending item more than once.
Limited project visibility	There is no single view of “what is happening” or “what needs my decision today.”

“The process is not lacking expertise; it is lacking integration.”

The BuildOS opportunity

BuildOS is the connective layer this journey is missing: a shared workspace where stakeholders, decisions, quotations, budget, documents and progress live in one place — with AI surfacing what needs the homeowner's attention instead of leaving them to reconstruct it manually. BuildOS augments the humans already doing the expert work; it does not replace them.

SECTION 03

Research Approach & Evidence Status

BuildOS is currently a product concept and portfolio proof-of-work, not a live company with customers. This case study is built on three completed source materials — a TAM/SAM/SOM market analysis, a user research guide, and initial user personas — combined with product management reasoning for the sections no source material yet exists for. Reading it accurately depends on knowing which claims fall into which category.

Category	What it means	How it's marked in this document
A — Validated / source-derived	Explicitly supported by the supplied TAM/SAM/SOM analysis, personas, or research framework.	Cited directly, with source and figure.
B — Proposed product strategy	Recommendations, prioritization, features, architecture, business model and roadmap developed in this case study.	Framed as “recommended” / “proposed.”
C — Assumption / hypothesis	A reasonable hypothesis not yet validated through interviews or a pilot.	Framed as “hypothesis” / “to validate.”

What has and hasn't happened

User interviews have been designed but not yet conducted. The personas below are product hypotheses built from the research framework's target segments, not findings from completed interviews. No claim in this document should be read as “research revealed...” — the accurate framing throughout is “initial hypotheses suggest...”, to be confirmed through the 12–15 interviews the guide below specifies.



PLAN / COORDINATE / BUILD / TOGETHER

USER RESEARCH GUIDE FOR BuildOS

Understand homeowners. Uncover real problems. Build a product that makes home construction simpler.

From Fragmented Construction
to a Connected Experience.

Build Smarter. Together.

The research framework used to plan (not yet execute) homeowner interviews — 12–15 one-on-one sessions, 45–60 minutes each, across define → discover → analyze → prioritize → apply.

SECTION 04

Market Opportunity — TAM / SAM / SOM

Market unit: number of new individual-house construction projects per year across four metros (Bengaluru, Chennai, Hyderabad, Pune), multiplied by a realistic platform revenue per project. Homeowners engage BuildOS once per house over a 12–24 month build, not as recurring subscribers — so the model prices a project-lifecycle fee rather than an annual subscription.

	Conservative	Base	Upside
TAM (annual)	₹21 crore	₹52 crore	₹104 crore
SAM (annual)	₹11 crore	₹28 crore	₹55 crore
SOM (Year-3 run-rate)	₹0.55 crore	₹2.2 crore	₹6.4 crore

TAM — bottom-up build

City	Metro population (2025)	Est. new-house projects/yr
Bengaluru	13.2M	~8,300 (directly reported)
Chennai	11.2M	~7,000 (population-scaled)
Hyderabad	9.2M	~5,800 (population-scaled)
Pune	6.8M	~4,300 (population-scaled)
Total	40.4M	~26,000 / year

Only Bengaluru has a directly reported figure (BBMP clears ~10,000 building-plan approvals/year, ~83% residential ≈ 8,300); the other three cities are extrapolated from Bengaluru's per-capita approval rate — a real limitation, flagged rather than hidden. Revenue per project has no public benchmark and is bracketed as a strategic assumption: ₹8,000 (conservative) / ₹20,000 (base) / ₹40,000 (upside).

Top-down cross-check: total individual-house construction spend across the four metros is estimated at ~₹14,000–16,000 crore/year. A 0.3–0.5% platform take-rate on that spend implies ₹42–70 crore/year — consistent with the bottom-up base case of ₹52 crore.

SAM — two filters narrow TAM

Filter	Effect	Basis
Business-model fit	–15%	Excludes homeowners already using full-turnkey/EPC builders (e.g. Brick & Bolt), who've outsourced the coordination problem to one vendor.
Digital readiness + willingness to pay	~60–65% of remainder	Urban India has ~71% internet penetration (IAMAI/Kantar, 2022); the weakest-sourced input in the model, since no survey data exists on willingness to pay for this category specifically.

Net: SAM lands at ~13,700–14,000 projects/year, roughly 53% of TAM.

SOM — Year-3 run-rate, built by city and rollout year

City	SAM (projects/yr)	Yr-3 penetration (base)	Projects captured
Bengaluru (Yr 1)	~4,400	15%	660
Chennai (Yr 2)	~3,700	6%	222
Hyderabad (Yr 2)	~3,050	5%	153

City	SAM (projects/yr)	Yr-3 penetration (base)	Projects captured
Pune (Yr 3)	~2,320	2%	46
Total			~1,080 / year

Sanity check: Brick & Bolt — a higher-friction, full-execution business — reported ~260 cumulative projects in Bengaluru after ~2 years; a Year-3 run-rate of 660/year for a lower-friction coordination product is plausible without being aggressive. By Year 5, as the other cities mature, SOM could reasonably run 2–4x higher.

KEY INSIGHTS

- The funnel decay is the point, not a weakness — TAM→SAM→SOM base case shrinks from ₹52cr → ₹28cr → ₹2.2cr; a lazier “1% of TAM” shortcut would have overstated Year-3 revenue by roughly 20x.
- Revenue-per-project design is the biggest single lever on SOM — it alone swings the outcome 6x between conservative and upside, meaning the monetization model deserves as much attention as acquisition.
- Brick & Bolt validates willingness-to-pay for construction-tech but also defines the differentiation risk: BuildOS must read as “coordinate your own chosen team,” not a lighter version of an existing full-service category.
- A real pilot beats further modeling — several inputs here rest on judgment calls that a paid Bengaluru pilot would resolve faster than refining the spreadsheet.

Major limitations: three of four city project counts are population-scaled proxies, not measured; unpermitted construction is excluded (likely understating the pool); revenue-per-project is unvalidated against real pricing research; and competitive response from incumbents is not modeled. Full source list: BBMP building-plan data (2024), Worldometers/UN metro estimates (2025), IMARC India Residential Construction report (2025, context only), Brick & Bolt company disclosures (2020–2025), IAMAI/Kantar Internet in India (2022).

SECTION 05

Competitive Landscape

Competitor analysis was not part of the supplied source material; the summary below reflects current publicly available information (September 2026), prioritizing India while using global tools as benchmarks. Claims are marked by evidence type — verified capability, observed market pattern, or strategic inference — rather than presented as uniformly confirmed.

Direct: full-service, turnkey construction

Player	What it verifiably offers	Evidence type
Brick & Bolt (India)	End-to-end turnkey home construction with fixed-price packages, escrow-based milestone payments, a published quality-check system, and a digital cost calculator; operates in 10+ Indian cities.	Verified (company site)
Similar EPC/turnkey builders	Outsource the entire coordination problem to one vendor rather than helping homeowners coordinate their own chosen team.	Observed market pattern

These are BuildOS's closest substitute, not its closest analogue: a homeowner who already hands the whole build to one EPC company doesn't need a coordination layer — this is the basis for the SAM's 15% business-model-fit exclusion.

Adjacent: construction management software (largely contractor-facing)

Player	Focus	Evidence type
BuildNext	India-built BOQ, estimation and site-progress tool for contractors/builders, priced for mid-size teams.	Verified (product listing)
Powerplay, FalconBrick, Onsite, RDash	India-market site/project tracking, budgets and reporting aimed at builders, PMCs and developers replacing WhatsApp/Excel internally.	Verified (product listings)
Buildertrend, Houzz Pro, CoConstruct, Procore	Global project-management platforms for contractors with client-facing portals; Houzz Pro reports reach across 3M+ professionals and 70M+ homeowners.	Verified (company sites)

These tools are built contractor-first, with a homeowner “view” bolted on as a portal. BuildOS inverts that: the homeowner is the primary user and owner of the workspace, with the contractor as an invited collaborator on their data.

Table-stakes vs. whitespace

TABLE-STAKES (EXPECTED BY 2026)

- Project dashboard / progress tracking
- Document and photo storage
- Budget vs. actual tracking
- Mobile-first site updates

BUILDOS WHITESPACE

- Homeowner-owned workspace, not a contractor tool with a client portal bolted on
- AI-assisted quotation comparison across multiple vendors, not just single-vendor tracking
- Cross-stakeholder coordination (architect + contractor + supplier + bank) in one shared context
- A path from assistance to human-approved agentic follow-ups (vendor chasing, budget alerts)

SECTION 06

User Personas

These are product personas / hypotheses built from the research framework's target segments — not findings from completed interviews. They should be validated against the 12–15 planned homeowner interviews before being treated as confirmed.

CASE STUDY

USER PERSONA

BuildOS
An Intelligence platform for Home Construction

PRIMARY USER

Ananya Rao

Homeowner & Project Owner

"I'm building a home for my family, and I just want everything to be organized, transparent and on track."

A home is a big dream. I just want to build it with confidence.

SECONDARY USER

Ravi Kumar

Contractor

"I want clear requirements, quick approvals and smooth communication so I can focus on delivering quality work."

A smarter way to build home.

Demographics

- Age range: 30 – 40
- Location: Hyderabad (Urban)
- Occupation: Product Manager (IT)
- Income band: ₹25 – 50 lakhs (per annum)
- Relationship: Married, 1 child
- Digital literacy: High

Psychographic

Pains

- Too many stakeholders and scattered information
- Unclear costs, timelines and progress
- Limited construction expertise

Gains

- Clarity and control over the project
- Confidence in decisions
- A smooth and transparent experience

Motivations

- Build a safe, quality home for family
- Make informed financial decisions
- Save time and reduce stress

Values

- Transparency
- Quality and reliability
- Long-term value

Fears

- Cost overruns
- Delays
- Compromised quality

Tech Proficiency

- Devices: Smartphone, Laptop
- Operating system: Android, macOS
- App / digital habits: WhatsApp, Gmail, Google Drive, Excel/Sheets, banking apps
- Communication platforms: WhatsApp, Email, Calls, Google Meet
- Internet constraints: None significant

Product-Specific

- Current solutions / alternatives: WhatsApp, spreadsheets, email, paper records
- Usage scenarios: Track progress, compare quotes, manage payments, store documents
- Expected usage frequency: Daily / several times a week
- Purchase / adoption drivers: Organization, transparency, better decisions
- Current satisfaction level with alternatives: Low
- Switching barriers: Habit, data migration effort, trust
- Success metrics: Project on time, within budget, with quality

What they want

- Have a clear view of project status at a glance.
- Compare quotations and options easily.
- Keep track of spending and upcoming payments.
- Get timely updates without chasing people.
- Make confident decisions with the right information.

Demographics

- Age range: 35 – 50
- Location: Hyderabad (Urban)
- Occupation: Independent Contractor
- Income band: ₹15 – 40 lakhs (per annum)
- Relationship: Married, with children
- Digital literacy: Medium

Psychographic

Pains

- Unclear or changing requirements
- Delays in approvals and payments
- Miscommunication across stakeholders

Gains

- Clear scope and timelines
- Faster decisions and payments
- Professional and organized collaboration

Motivations

- Complete projects on time
- Maintain good client relationships
- Grow reputation and business

Values

- Trust and transparency
- Quality workmanship
- Long-term client relationships

Fears

- Payment delays
- Scope creep
- Negative reviews or reputation damage

Tech Proficiency

- Devices: Smartphone, Laptop (occasionally)
- Operating system: Android, Windows
- App / digital habits: WhatsApp, Google Drive, YouTube (for references), UPI apps
- Communication platforms: WhatsApp, Calls
- Internet constraints: Occasional connectivity issues on site

Product-Specific

- Current solutions / alternatives: WhatsApp, calls, paper notes
- Usage scenarios: Share progress updates, receive approvals, manage tasks and payments
- Expected usage frequency: Daily
- Purchase / adoption drivers: Easier communication, faster approvals, organized documentation
- Current satisfaction level with alternatives: Medium
- Switching barriers: Time to learn, habit, perceived complexity
- Success metrics: Smooth project execution, timely payments, satisfied clients

What they want

- Receive clear, well-documented requirements.
- Share site updates easily.
- Get faster approvals and payments.
- Keep all project communication in one place.
- Build a strong track record with satisfied homeowners.

PRODUCT IMPLICATIONS

- Reduce coordination overhead for homeowners and contractors.
- Surface decisions and approvals that need homeowner attention.
- Maintain a shared source of truth across all stakeholders.
- Support simple, mobile-first usage for on-site updates.
- Drive transparency in timelines, spending and progress.

Better Homes. Brighter Futures.

Primary — Ananya Rao, Homeowner & Project Owner

Dimension	Detail
Context	30–40, Hyderabad (urban), Product Manager (IT); building a family home with high financial and emotional investment and limited construction expertise.
Jobs-to-be-done	Keep the project organized and on track; compare vendor quotations confidently; make informed financial decisions without becoming a construction expert.
Pain points	Too many stakeholders and scattered information; unclear costs, timelines and progress; limited domain expertise to evaluate decisions.
Current tools/workarounds	WhatsApp, spreadsheets, email, phone calls, paper records.
Digital behaviour	High digital literacy; daily/several-times-a-week usage expected of a project tool; comfortable across smartphone and laptop.
Adoption barriers	Habit, data-migration effort into a new tool, and trust that the platform reflects reality accurately.
Desired outcome	Clarity and control over the project; confidence in decisions; a smooth, transparent build.

Secondary — Ravi Kumar, Contractor

Dimension	Detail
Context	35–50, Hyderabad (urban), independent contractor managing multiple client relationships and crews.
Jobs-to-be-done	Complete projects on time; maintain client relationships; keep scope and payment terms unambiguous.
Pain points	Unclear or changing requirements; delays in approvals and payments; miscommunication across stakeholders.
Current tools/workarounds	WhatsApp, calls, paper notes.
Digital behaviour	Medium digital literacy; daily usage expected for sharing site updates; occasional connectivity constraints on-site.
Adoption barriers	Time to learn a new tool, habit, and perceived complexity relative to WhatsApp.
Desired outcome	Clear, well-documented requirements; faster approvals and payments; a stronger track record with satisfied homeowners.

WHAT EACH PERSONA NEEDS BUILDOS TO DO

- Ananya needs one place that shows project status at a glance, lets her compare quotations side-by-side, and tells her what needs her decision today — without requiring her to become a construction expert.
- Ravi needs unambiguous, well-documented requirements and approvals, a simple way to push site updates from his phone, and faster, more predictable payments — without adding administrative overhead to his day.

SECTION 07

Current User Journey

The homeowner's build runs through ten stages. Fragmentation isn't uniform across them — some stages carry far more coordination burden than others, and those are where BuildOS should concentrate first.

Stage	User goal	Stakeholders	Current tools	Friction	BuildOS opportunity
Planning	Define scope, budget, plot	Homeowner, architect	Notes, conversations	No shared reference point for scope or budget from day one	Establish the project workspace and budget baseline
Vendor discovery	Find architects, contractors, suppliers	Homeowner	Referrals, calls, site visits	No structured way to compare vendors on the same criteria	Structured vendor profiles and comparison
Quotations	Collect & compare quotes	Homeowner, vendors	WhatsApp, PDFs, spreadsheets	Manually re-typing quotations to compare cost and scope — highest-friction stage	AI-assisted quotation extraction & comparison
Budgeting	Set and track the budget	Homeowner	Spreadsheets, bank statements	Spend-to-date reconstructed after the fact, not tracked live	Live budget tracking with deviation alerts
Design/approval	Approve drawings & changes	Homeowner, architect	Email, printed drawings	Approval history scattered across email threads	Centralized approval log with version history
Procurement	Order & schedule materials	Homeowner, contractor, suppliers	Calls, WhatsApp	Material timing disconnected from construction schedule	Material planning linked to project timeline
Construction	Execute the build	Contractor, crew	Site visits, photos	Progress visibility depends on the homeowner physically visiting	Structured progress updates & photo log
Progress tracking	Monitor milestones	Homeowner, contractor	Photos, calls	No single dashboard of milestones vs. plan	Milestone dashboard visible to all stakeholders
Payments	Release payments to vendors	Homeowner, contractor, bank	Bank transfers, receipts	Payment status disputes from missing documentation	Payment tracking tied to milestone completion
Handover	Close out & document	Homeowner, contractor, architect	Paper records	No consolidated project history to reference later	Complete, exportable project record

HIGHEST-BURDEN MOMENTS

- Quotations — the most manually intensive, highest-decision-stakes stage, and the strongest MVP entry point.
- Budgeting — opacity here compounds every later decision, since every subsequent choice is made against an unclear baseline.
- Progress tracking — the stage where “what is happening right now” is least visible to the person paying for it.

SECTION 08

Product Vision & Concept

VISION

A residential construction journey where every stakeholder works from the same information, and the homeowner always knows what's happening and what needs their decision.

VALUE PROPOSITION

BuildOS replaces the homeowner's fragmented WhatsApp-and-spreadsheet coordination with one connected workspace and AI that surfaces the decisions and risks that matter.

PRODUCT PRINCIPLES

- Context before intelligence — AI recommends only once structured data exists.
- Human expertise stays authoritative — BuildOS augments, never replaces, professionals.
- Approval before action — consequential steps always require homeowner sign-off.

NORTH-STAR EXPERIENCE

A homeowner opens BuildOS once and can answer, in under a minute: what happened, what's pending, and what needs my decision today.

BuildOS as an intelligent coordination layer

1 Fragmented information	2 Manual coordination	3 Reactive decisions	4 Limited visibility
TODAY → BUILDOS			
1 Connected information	2 Contextual intelligence	3 Proactive insights	4 Coordinated action
WHAT BUILDOS DOES NOT REPLACE Architects, structural engineers, contractors, and other domain experts remain the authority on design, engineering and execution decisions. BuildOS connects their work and outputs into a shared context — it does not perform construction, design, or engineering judgment on their behalf.			

SECTION 09

MVP & Feature Prioritization

Capabilities are evaluated on user value, business value and feasibility, then sorted with MoSCoW. The MVP is deliberately narrow: it establishes the shared workspace and solves the two highest-friction stages (quotations and budget) before any AI feature is introduced.

Priority	Capability	Why
Must have	Project workspace & dashboard	The foundational shared source of truth every other feature depends on.
Must have	Stakeholder management	Without a structured stakeholder list, nothing else can be attributed or routed correctly.
Must have	Quotation collection & comparison	The highest-friction, highest-decision-stakes stage identified in the journey map.
Must have	Budget estimation & tracking	Budget opacity compounds every later decision; must be live, not reconstructed.
Must have	Tasks & milestones	Gives the dashboard something concrete to show progress against.
Must have	Document management	Quotations, approvals and receipts need one home, not scattered email threads.
Should have	Site updates & photos	High value, but can launch as a simple upload before richer progress analytics.
Should have	Payment tracking	Important, but can follow milestone tracking rather than launch alongside it.
Should have	Notifications & alerts	Meaningfully improves engagement but is additive to a working core workspace.
Could have	AI decision support (quotation extraction, budget-deviation flags)	High value, but depends on Must-have data existing first — sequenced right after MVP core, not inside it.
Could have	Vendor discovery & marketplace	Valuable later, but requires a supply-side network BuildOS doesn't have at launch.
Won't have (for now)	Agentic workflows (automated vendor follow-ups, autonomous actions)	Requires trust and data maturity the MVP hasn't yet earned; sequenced into Phase 3.

AI is introduced selectively: quotation comparison and budget-deviation detection are the first candidates because the MVP's Must-have data (quotations, budget entries) already provides the structured context they need — consistent with the “context before intelligence” principle.

SECTION 10

SCAMPER Analysis

Applied directly to the BuildOS MVP, not as a generic framework exercise — each row challenges a real construction workflow and proposes a specific product response.

Lever	Existing workflow	BuildOS improvement	User value
Substitute	WhatsApp for project coordination	Replace ad-hoc chat threads with a structured project feed tied to stakeholders and decisions	Nothing gets buried in an unrelated chat history; anyone can find the current state instantly
Combine	Quotation collection + budget tracking (today: separate spreadsheets)	Combine quotation comparison and budget impact into one view, so approving a quote updates the budget automatically	One decision, one action — no manual re-entry between tools
Adapt	Milestone-based escrow used by turnkey builders (e.g. Brick & Bolt)	Adapt milestone-linked payment tracking for homeowners who've assembled their own team, not a single EPC vendor	Payment transparency without giving up control of vendor choice
Modify	Manual, reactive vendor follow-ups	Modify follow-ups from homeowner-initiated to system-prompted, with a suggested message ready for approval	Fewer forgotten follow-ups without adding the homeowner's mental load
Put to another use	Site photos, normally just a status update	Reuse the same photo stream as a timestamped visual project history for future reference or disputes	A ready-made record for warranty claims, resale, or disagreements
Eliminate	Re-typing the same information across quotation, budget and payment records	Eliminate duplicate data entry by making the quotation the single source that budget and payment both read from	Less admin work; fewer transcription errors
Reverse / Rearrange	Homeowner chases contractor for updates	Rearrange the default: the system prompts the contractor for a scheduled update rather than waiting to be asked	Shifts the coordination burden off the least-informed party

MOST VALUABLE OPPORTUNITIES

- Combine quotation comparison with live budget impact — turns two disconnected spreadsheets into one decision moment.
- Eliminate duplicate data entry across quotation, budget and payment records.
- Reverse the follow-up default so the system, not the homeowner, initiates the nudge.

SECTION 11

AI & Intelligence Layer

AI in BuildOS follows a consistent progression from raw data to an approved action — it is never a standalone chatbot layered on top of an otherwise unconnected product.

1 Data	2 Context	3 Intelligence	4 Recommendation	5 Human approval	6 Action	7 Feedback
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Capability	Input	AI processing	Output	Human oversight	MVP / Future
AI project summarization	Tasks, updates, documents this week	Synthesize into a short status narrative	“What changed this week” summary	Homeowner reviews; not a substitute for reading key documents	MVP+
AI decision copilot	Pending decisions & related context	Surface the relevant facts (cost, timeline, past decisions) for a choice	A framed decision with trade-offs, not a directive	Homeowner makes the final call	Future
Quotation extraction & comparison	Uploaded vendor quotation (PDF/image)	Extract line items, cost, scope into structured fields	Side-by-side comparison table across vendors	Homeowner confirms extracted data is accurate before it's used	MVP
Budget anomaly detection	New quotation or expense entry	Compare against approved budget line	Flag when a cost deviates beyond a threshold	Homeowner reviews the flag and decides the response	MVP
Material recommendation	Project specs, past material choices	Match specs to viable material/vendor options	Ranked suggestions with rationale	Contractor/architect input required before purchase	Future
Timeline risk detection	Task dates vs. dependencies	Detect schedule slippage patterns	Early warning on at-risk milestones	Homeowner and contractor jointly review	Future
Vendor follow-up assistance	Quotation pending beyond expected response time	Draft a follow-up message and suggest timing	Ready-to-send follow-up for approval	Homeowner approves before it's sent	Future (agentic)
Intelligent alerts	Any tracked threshold (budget, schedule, approval)	Prioritize what needs attention now	Ranked notification feed	Homeowner decides what to act on	MVP
Document understanding	Uploaded contracts, approvals, permits	Extract key terms and obligations	Searchable, structured document index	Legal/contract terms still reviewed by the homeowner or advisor	Future
Future autonomous workflows	Multiple linked signals across the project	Chain several of the above into a single workflow	A recommended multi-step action plan	Human approval remains required for any binding step	Future (agentic)

AI PRODUCT PRINCIPLES

— Context before intelligence; human expertise remains authoritative; human-in-the-loop for consequential actions.

Capability	Input	AI processing	Output	Human oversight	MVP / Future
— Explainable recommendations with visible source/context; confidence-aware, never presented as certain. — Privacy and role-based access, auditability, and guardrails against hallucination throughout. — Gradual progression from assistance to autonomy — never a jump straight to unsupervised action.					

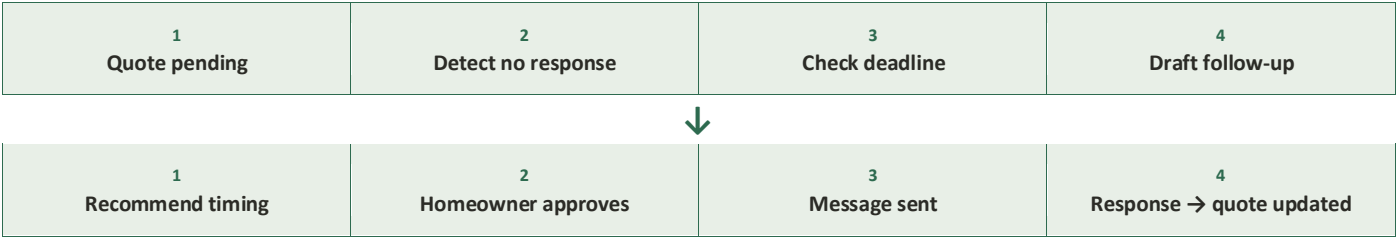
SECTION 12

Agentic AI Opportunity

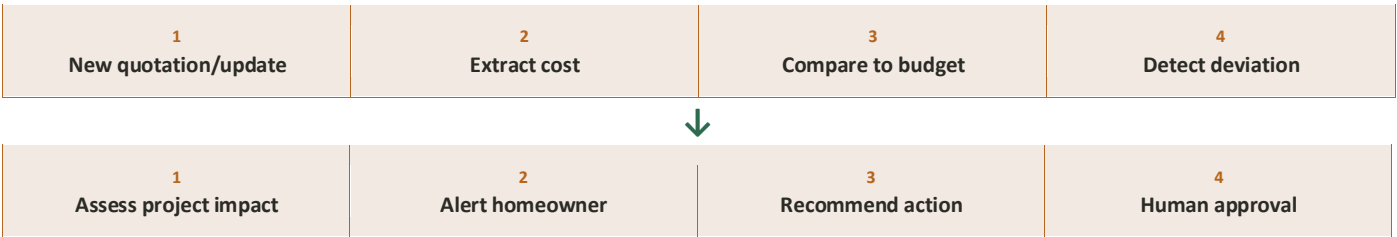
Once the MVP has established structured, trusted data, specific workflows can graduate from assistance to agentic execution — always with human approval retained for the consequential step.

	AI Assistant	AI Agent
What it does	Surfaces information and a recommendation; the human decides and acts	Carries out multi-step work on the human's behalf, pausing for approval at consequential points
Example	Flags that a quotation looks high relative to budget	Drafts and sends a vendor follow-up once approved, then updates the quotation record with the response
Risk profile	Low — output is informational	Higher — requires guardrails and an explicit approval gate before any binding action

Workflow 1 — Vendor Follow-up Agent



Workflow 2 — Budget Risk Agent



In both workflows, every step up to and including the recommendation runs autonomously; the step that commits money, sends a message, or changes a record on someone else's behalf always waits for explicit homeowner approval.

SECTION 13

Product & System Architecture

A layered architecture keeps the AI and integration layers replaceable without disturbing the core data model — important given how early-stage and unproven several external dependencies (vendor APIs, payment rails) are for this category.

User Experience

Web application · Mobile experience

Application Layer

Project management · Stakeholder management · Budget · Procurement · Documents · Notifications · Collaboration

Data Layer

Project data · Stakeholder data · Quotations · Budget / financial records · Documents · Activity history

AI / Intelligence Layer

LLM · Retrieval / knowledge layer · Document intelligence · Recommendation engine · Agent orchestration · Rules / validation · Guardrails

Integration Layer

Communication · Payments · Maps / location · Notifications · External vendor / service APIs

Security & Governance

Authentication · Role-based access · Data privacy · Audit logs · AI guardrails · Human approval

Data flow: user actions in the experience layer write to the application layer, which persists to the data layer. The AI layer reads from the data layer through the retrieval/knowledge component — it never writes directly; all AI outputs pass through rules/validation and guardrails, then require human approval before the application layer commits any resulting change. The security and governance layer wraps every other layer, controlling who can see and approve what.

SECTION 14

Technology & Implementation Approach

Technology choices are evaluated on speed to MVP, scalability, security, AI integration, developer ecosystem, maintainability and cost — not selected for their own sake.

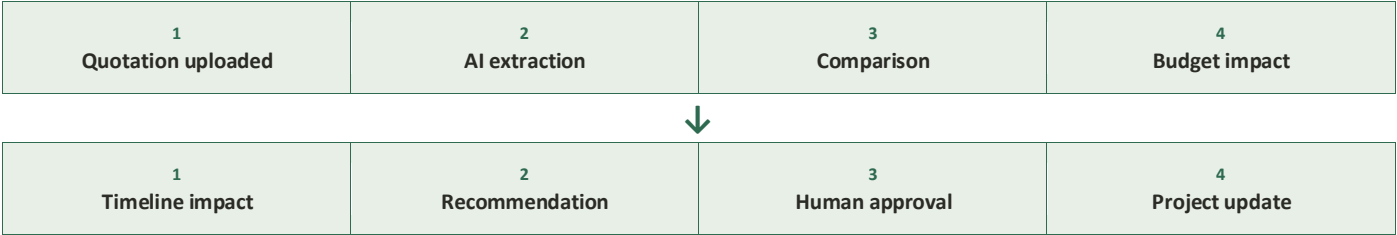
Layer	Prototype stack	Production-ready stack
Frontend	Lovable / low-code builder for rapid UI iteration	React / Next.js
Backend	Supabase edge functions / no-code workflow	Python (FastAPI) or Node.js services
Database	Supabase (managed Postgres)	PostgreSQL, self-managed or managed cloud
Authentication	Supabase Auth	Supabase Auth or a dedicated identity provider (e.g. Auth0)
File / document storage	Supabase Storage	Cloud object storage (e.g. S3-compatible)
AI / LLM	Claude or ChatGPT via API, prompted directly	LLM API with a structured orchestration layer and guardrails
Vector search / retrieval	pgvector on Supabase Postgres	pgvector or a dedicated vector database at scale
Workflow orchestration	n8n / Make for prototyping agent-style flows	Dedicated orchestration service with approval gates
Notifications	Third-party notification API	Same, integrated with the application layer's event system
Analytics	Product analytics tool (e.g. PostHog, Mixpanel)	Same, extended with a data warehouse for reporting
Hosting / cloud	Vercel + Supabase	Cloud provider (AWS/GCP/Azure) with managed Postgres

Tools such as Lovable, n8n, Make, Google AI Studio, Claude and ChatGPT are named here as prototyping and development accelerators for reaching an MVP quickly — not as an assumed final production architecture, which should be re-evaluated against real usage, scale and cost once the MVP validates demand.

SECTION 15

End-to-End AI Workflow — Proof of Concept

The strongest proof-of-concept workflow in BuildOS: a homeowner uploads a vendor quotation, and the platform turns it into a budget- and timeline-aware recommendation before anything changes in the project record.



Step	What happens
Trigger	Homeowner uploads a quotation (PDF or photo) for a pending vendor decision.
Data ingestion	The document is stored and queued for extraction; the raw file remains the source of truth.
AI processing	The model extracts line items, total cost, scope and validity period into structured fields.
Validation	Extracted data is checked against expected fields and flagged if confidence is low or fields are missing.
Recommendation	The system compares the quotation to the approved budget line and to any other quotations for the same scope, and generates a plain-language recommendation with its reasoning shown.
Human approval	The homeowner reviews the extracted data and recommendation, corrects anything wrong, and approves or rejects.
Action	On approval, the budget record and vendor status update automatically; on rejection, nothing changes and the homeowner can request more information.
Feedback loop	Corrections the homeowner makes to extracted data are logged to improve extraction confidence over time.

SECTION 16

Product Roadmap & Project Plan

Phase	Focus	Key deliverables
Phase 0 — Discovery & Validation	De-risk the core assumptions before building	12–15 homeowner interviews; problem validation; a clickable prototype of the quotation/budget flow
Phase 1 — MVP	The connected workspace	Project workspace, stakeholders, quotations, budget, tasks, documents, progress dashboard
Phase 2 — Intelligence	AI decision support where context now exists	AI project summaries, quotation extraction/comparison, budget anomaly detection, smart alerts
Phase 3 — Automation	Assistance graduates to agentic, human-approved workflows	Vendor follow-up agent, payment workflows, schedule risk detection, procurement automation
Phase 4 — Ecosystem	Extend beyond the single build	Supplier and financial-institution partnerships, professional network, advanced agentic workflows

Initial MVP sprint plan (illustrative, 12 weeks)

Sprints	Focus
1–2	Project workspace shell, authentication, roles (homeowner / contractor / architect)
3–4	Stakeholder management and project dashboard
5–6	Quotation upload, manual comparison view; document storage
7–8	Budget entry and tracking; tasks & milestones
9–10	Site updates & photo uploads; notifications
11–12	Internal QA, pilot onboarding with a small cohort of Bengaluru homeowners, feedback loop

SECTION 17

Business Model & Go-to-Market

Monetization options considered

Model	Fit for BuildOS
Homeowner subscription	Weak fit — a 12–24 month one-time build doesn't match a recurring-subscription mental model.
Freemium → paid	Reasonable for acquisition; core dashboard free, quotation/budget intelligence paid.
Project-based fee	Strong fit — matches the TAM/SAM/SOM model's own unit of a per-project fee, and matches how homeowners already think about one-time build costs.
Transaction / commission	Possible longer-term (e.g. on vendor payments), but adds trust friction early and isn't needed for MVP viability.
Vendor / professional subscription	Viable secondary stream once BuildOS has homeowner-side scale and data worth paying to reach.
B2B2C via partnerships	Strong longer-term channel (architects/contractors bringing homeowners onto BuildOS) but requires proven homeowner value first.
Marketplace revenue	Requires supply-side density BuildOS won't have at launch; a Phase 4 consideration.

RECOMMENDED INITIAL MODEL

A project-lifecycle fee paid by the homeowner (flat or value-tiered), consistent with the ₹8k–₹40k per-project revenue bands used in the TAM/SAM/SOM model. Homeowners pay once for coordination clarity across their build; recurring revenue and network effects are earned later through vendor/professional subscriptions and partnerships once BuildOS has demonstrated homeowner-side value.

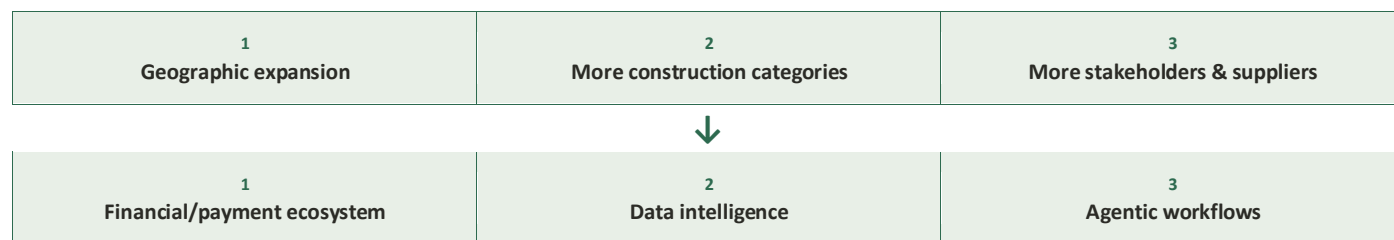
Go-to-market: initial channel

Architect and contractor referral partnerships in Bengaluru are the recommended first channel — homeowners trust their architect/contractor's recommendation at exactly the moment they're assembling a team, and CAC through broad paid marketing is high for a one-time, high-consideration purchase. Homeowner communities and word-of-mouth referrals are secondary channels once a pilot cohort exists; other channels (material suppliers, consultants, broad partnerships) are deferred until the primary channel is proven.

SECTION 18

Scalability, Metrics & Future Vision

A — Scale



PRODUCT & TECHNICAL SCALABILITY

- The layered architecture (Section 13) lets new cities and stakeholder types plug into the existing data model without redesigning the core.
- AI components (extraction, comparison) generalize across cities once trained on structured quotation/budget data
 - the harder scaling problem is data quality, not model capability.

ECOSYSTEM & DATA NETWORK EFFECTS

- Each completed project adds structured data (typical costs, timelines, vendor reliability) that improves recommendations for the next homeowner in the same city.
- A large enough base of trusted project data becomes valuable to suppliers and lenders, opening the B2B2C and marketplace paths deferred from the initial model.

B — Success metrics

NORTH STAR METRIC

Share of active projects with an up-to-date budget and at least one AI-assisted decision per month — a single metric that captures both engagement and whether BuildOS is doing its actual job (coordination clarity), not just logins.

Category	Metric
Activation	Homeowner completes project setup and invites at least one stakeholder
Engagement	Weekly active use of the dashboard by the homeowner
Retention	Project remains actively updated through to handover, not abandoned mid-build
Decision completion	Share of pending decisions resolved within the platform vs. off-platform
Time saved	Reduction in time-to-decision on a quotation, self-reported and platform-timed
Vendor response rate	Share of follow-ups that receive a vendor response within the expected window
Budget visibility	Share of projects with budget updated within the last 7 days
AI recommendation acceptance	Share of AI recommendations approved as-is vs. edited or rejected
Project completion	Share of onboarded projects that reach handover on BuildOS
Revenue	Realized revenue per completed project vs. the ₹8k–₹40k modeled bands

C — Future vision

1 Connect	2 Understand	3 Predict	4 Recommend	5 Act
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TODAY

Construction coordination platform

TOMORROW

Construction intelligence platform

LONG-TERM

An intelligent construction operating system / ecosystem — the connective and data layer for how residential construction gets planned, funded, built and handed over.

FINAL PAGE

Product Thesis

Question	Answer
Why BuildOS?	Because the problem homeowners face isn't a lack of expertise — it's the absence of a shared source of truth across the people they've hired. That's a coordination problem, and coordination problems are what software is good at solving.
Why now?	Digital adoption among urban Indian homeowners is high enough to support a software-first workflow, while the individual-home construction market itself remains genuinely under-digitized — the gap between those two facts is the opportunity.
Why this user?	The homeowner carries the highest financial and emotional stake in the project and the least expertise to manage it alone — they are the party with the most to gain from a coordination layer, not the professionals who already have their own tools.
Why this approach?	Starting narrow — one workspace, two highest-friction stages — earns trust before asking for more; a feature-complete platform on day one would dilute focus and slow the very validation this concept still needs.
Why AI?	Because the raw inputs already exist (quotations, updates, budgets) but nobody has time to structure them by hand; AI's job is to turn scattered inputs into a decision a homeowner can act on, with a human approving every consequential step.
Why can it scale?	Each completed project adds structured data that makes the next homeowner's recommendations better — the same mechanism that, over time, opens a path to suppliers, lenders and the wider professional ecosystem.

***BuildOS does not replace the people who build homes.
It connects the information, decisions and actions that let them build better.***